

Chapter 8

Welfare, Monopoly, and Price Discrimination

"Competition is the keen cutting edge of business, always shaving away at costs."

— Henry Ford

Connecting Week 7 to Week 8: From How Much to How Good

Week 7 Established

How a competitive firm maximises profit: set $P = MC$. The supply curve emerges from the SMC curve above $\min(SAVC)$. Short-run market equilibrium where $Q^s = Q^d$.

Left open: Is this outcome actually *good* for society?

How does it compare to other market structures?

Week 8 Answers It

We measure "good" with total surplus $TS = CS + PS$. Competitive equilibrium *maximises* TS . Monopoly restricts output below the competitive level, creating deadweight loss $DWL = TS_{PC} - TS_M$. Price discrimination can eliminate DWL but extracts all consumer surplus.

The central question this week: not just "what does the market do?" but "is what it does efficient and equitable?"

Before You Begin: Quick Self-Assessment

Test yourself on these prerequisites before reading:

1. **What is the competitive firm's profit-maximising rule?** *Produce where $P = MC$; shut down if $P < AVC$. From Chapter 7.*
2. **What is consumer surplus? How is it measured?** *The difference between WTP and price paid; the area under the demand curve above the price line.*
3. **How does a monopolist differ from a competitive firm in its pricing decision?** *A monopolist faces a downward-sloping demand curve and sets $MR = MC$ where $MR < P$. A competitive firm takes price as given.*
4. **What is deadweight loss?** *The value of mutually beneficial trades that do not occur because of a market distortion — pure social waste, captured by nobody.*
5. **Give one example each of first-, second-, and third-degree price discrimination.** *First: personalised pricing (car dealerships). Second: quantity discounts or versioning. Third: student/senior discounts.*

Questions This Chapter Will Answer

- How do we measure whether a market outcome is good for society — in monetary terms, regardless of what is being traded?
- Why does restricting output below the competitive level destroy value that no one captures?
- Why do cheaper imports hurt domestic producers but benefit the country as a whole?
- A monopolist charges \$7 for something that costs \$4 to make. Who loses and how much?
- Can a firm ever increase efficiency by charging different consumers different prices?
- What is the trade-off between efficiency and equity when a firm price discriminates perfectly?

Chapter Road Map. Chapter 8 begins with a monetary yardstick for evaluating market outcomes — total surplus ($TS = CS + PS$). We show that the competitive equilibrium maximises TS and any deviation creates deadweight loss. We then apply the framework to policy: opening trade raises TS while a tariff creates two DWL triangles. The second half of the chapter studies monopoly: a single seller restricts output below the competitive level, raising price and generating a DWL triangle. Finally, we examine price discrimination — how a monopolist can use information about buyer willingness to pay to extract additional surplus. The chapter closes with a welfare table comparing competition, uniform monopoly, and perfect price discrimination on both efficiency and equity.

Learning Objectives and Topics

Each objective states what you will be able to *do*.

LO1 Compute consumer and producer surplus from linear supply and demand curves.

§8.2–8.3 *CS and PS*

LO2 Define economic efficiency as TS maximisation; identify and measure deadweight loss.

§8.4 *Total Surplus & Efficiency*

LO3 Apply the surplus framework to trade and tariff policy; calculate gains from trade and tariff DWL.

§8.5 *Policy Interventions*

LO4 Find the monopoly optimum using the two-step rule ($MR = MC$, then read demand); measure monopoly profit.

§8.6–8.8 *Monopoly*

LO5 Compare competitive and monopoly outcomes using CS, PS, TS, and DWL; compute the Lerner Index.

§8.7 *Welfare Comparison*

LO6 Analyse perfect price discrimination and market separation; explain the efficiency–equity trade-off.

§8.9 *Price Discrimination*

Notation Used in This Chapter.

CS	consumer surplus	TS	total surplus = CS + PS
PS	producer surplus	DWL	deadweight loss
WTP	willingness to pay	P_M, Q_M	monopoly price, quantity
P_W, P_D	world, domestic price	P_{PC}, Q_{PC}	competitive price, quantity
MR	marginal revenue	L	Lerner Index = $(P - MC)/P$
t	tariff (specific)	ε_d	own-price elasticity of demand

8.1. Introduction: Comparing Market Outcomes

Markets for iPads and markets for wheat operate at radically different prices and serve very different purposes. How do we compare them on a common basis — and how do we know whether any given market outcome is “good”?

Maria Returns. Last week, Maria maximised honey-farm profit by setting $P^* = SMC$. This week we ask a different question: was the resulting market outcome *good for society*? We need a common measuring rod. Total surplus — the sum of what buyers and sellers gain from trade — is exactly that rod.

The Toolkit This Chapter Adds.

Concept	Measures	Used to evaluate
Consumer surplus (CS)	Buyer gains	How much consumers benefit
Producer surplus (PS)	Seller gains	How much producers benefit
Total surplus (TS = CS + PS)	Social value created	Whether an outcome is efficient
Deadweight loss (DWL)	Value destroyed	Cost of restricting output

Total surplus is expressed in dollars, making it a unit-free metric. Whether we are analysing iPads or apples, we can compute TS and compare market outcomes on a common scale.

Notation Used in This Chapter

CS	consumer surplus = $WTP - P$	TS	total surplus = CS + PS
PS	producer surplus = $P - MC$	DWL	deadweight loss
WTP	willingness to pay	Q^*, P^*	competitive equilibrium
P_W	world price	P_R	tariff-inclusive price
Q_M^*, P_M^*	monopoly output, price	L	Lerner Index
MR	marginal revenue = $a - 2bQ$	ε_d	price elasticity of demand

Why This Matters. Every policy debate about price controls, tariffs, monopoly regulation, or antitrust enforcement ultimately reduces to a welfare question: who gains, who loses, and what is the net effect on TS? Without this framework, we can only say “consumers lose” or “producers gain.” With it, we can quantify both effects and determine whether the overall outcome is better or worse.

We start by building the two components of total surplus: consumer surplus and producer surplus.

8.2. Consumer Surplus

You would pay \$1,000 for an iPad but only have to pay \$450. Are you \$550 better off than if you had to pay your full willingness? The answer — yes, by exactly \$550 — defines consumer surplus.

8.2.1. Definition

Consumer Surplus (CS):

$$CS = \text{Willingness to pay} - \text{Price paid} \quad (8.1)$$

In aggregate: the area under the demand curve and above the market price.

In words: CS is the “deal” a buyer gets. The demand curve traces out willingness to pay for each unit. CS is the area between the demand curve and the price line.

Campus Coffee Shop. Price: \$4/latte. Three students:

- Alice (WTP \$7): CS = \$3.00. ✓buys.
- Ben (WTP \$5.50): CS = \$1.50. ✓buys.
- Cara (WTP \$3.80): CS = −\$0.20. ✗does not buy.

Total CS = \$4.50. If price rises to \$5: Ben stops buying, CS falls to \$2.00. Both the number of buyers and each buyer’s surplus fall.

CS = area under demand, above price. Loses area when price rises; gains area when price falls.

Try It: Computing CS and PS. Suppose $D : P = 12 - Q$ and $S : P = 2 + Q$.

- Find Q^* and P^* .
- Compute CS as the area of a triangle: $CS = \frac{1}{2} \times \text{base} \times \text{height}$.
- Compute PS similarly. What is TS?
- If the price is fixed at \$9 (a price ceiling scenario): compute the new CS and PS. Has TS fallen?

Check Your Understanding

- iPad WTP \$1,000, price \$450. CS? If price falls to \$350, by how much does CS rise?
- A price rise from P_1 to P_2 on Q^* units. Using the trapezoid formula, express the

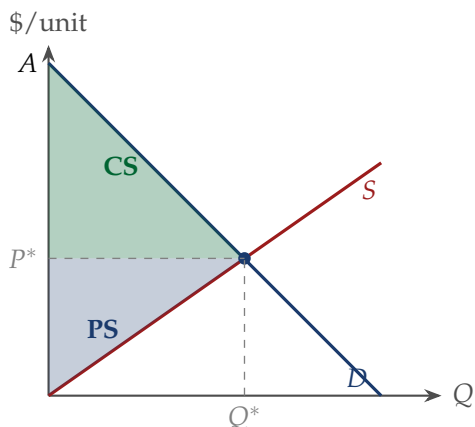


Figure 8.1. CS and PS at competitive equilibrium. What to look for: CS (green) is the triangle between the demand curve and P^* — the aggregate buyer gain. PS (blue) is the triangle between P^* and the supply curve — the aggregate seller gain. Together they form $TS = CS + PS$.

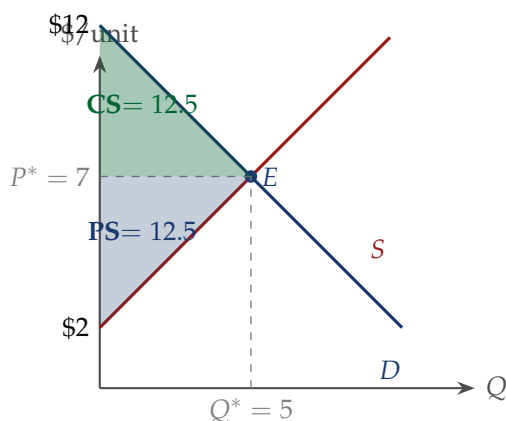


Figure 8.2. Consumer and producer surplus: $D : P = 12 - Q$, $S : P = 2 + Q$. What to look for: At equilibrium ($Q^* = 5$, $P^* = \$7$), CS (green) $= \frac{1}{2}(12 - 7)(5) = \12.5 and PS (blue) $= \frac{1}{2}(7 - 2)(5) = \12.5 . Total surplus $= \$25$. Both triangles have base $Q^* = 5$ and height equal to the gap between the market price and the curve intercept.

CS loss geometrically.

3. True or false: “A consumer who pays exactly their WTP earns zero consumer surplus.”

Consumer surplus captures buyer gains. Producer surplus captures seller gains — the mirror concept.

8.3. Producer Surplus

A baker’s marginal cost for a sourdough loaf is \$3. The market price is \$7. She earns \$4 on that loaf — more than she “needed” to produce it. This gap is producer surplus.

Producer Surplus (PS):

$$PS = \text{Price received} - \text{Willingness to sell (MC)} \quad (8.2)$$

In aggregate: the area above the supply curve and below the market price.

In words: PS is the “deal” a seller gets. The supply curve traces minimum acceptable prices (marginal costs). PS is the area between the price line and the supply curve.

PS $\neq$ Profit. Producer surplus = price minus marginal (variable) cost, represented by the area above the supply curve. Profit = revenue minus *total* cost (including fixed costs). When fixed costs are zero, PS = profit. When fixed costs are present, profit = PS – fixed costs. Always check whether fixed costs are present before equating them.

Consumer surplus is what you get for free. Producer surplus is what you charge for less than you could have. Both measure luck in a market — the difference between what you’d accept and what you actually receive.

Check Your Understanding

1. Supply is $P^s = 2 + Q$. Market price $P^* = \$7$, $Q^* = 5$. Compute PS as the area of the triangle.
2. A baker has fixed costs of \$100/day and $MC = \$3$ per loaf. Market price \$7, 200 loaves sold. Compute PS and profit. Why do they differ?
3. True or false: “A firm that earns zero economic profit has zero producer surplus.”

With CS and PS defined, we can define total surplus and economic efficiency.

8.4. Economic Efficiency and Total Surplus

The competitive equilibrium sets $P = MC$. Is this a coincidence — or is there something special about this output level that makes it the best the market can do?

8.4.1. Economic Efficiency

Economic Efficiency: An allocation is **economically efficient** if total surplus $TS = CS + PS$ is maximised. Equivalently: there are no remaining mutually beneficial exchanges — no buyer–seller pair exists where trade would make both better off.

In words: Efficiency means the total pie (CS+PS) is as large as possible. It says nothing about how the pie is divided.

8.4.2. Why Competitive Equilibrium Is Efficient

At (P^*, Q^*) , every unit for which $WTP > MC$ is traded, and no unit where $WTP < MC$ is produced. Both conditions hold simultaneously at the competitive equilibrium — which

Efficiency $\neq$ equity. A market can maximise TS while distributing it very unequally.

is exactly why it maximises TS.

8.4.3. Deadweight Loss

Deadweight Loss (DWL): The reduction in TS caused by a market distortion — an output level different from Q^* . DWL is the value of mutually beneficial trades that do not occur. It is **captured by nobody**: not consumers, not producers, not the government.

$$DWL = TS_{Q^*} - TS_{\text{actual}} \quad (8.3)$$

In words: DWL is not a transfer from one party to another. It is value destroyed outright.

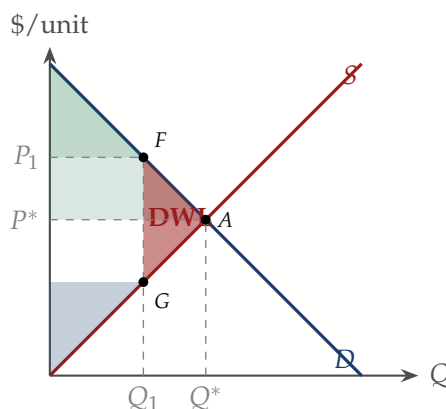


Figure 8.3. Deadweight loss from output restriction $Q_1 < Q^*$. *What to look for:* The red triangle $DWL = \text{area } FGA$ is the value of trades that could have occurred between Q_1 and Q^* (buyer $WTP >$ seller MC) but did not. This value is not captured by anyone — it is destroyed. At Q^* , $DWL = 0$ and TS is maximised.

Economic Insight. The same argument applies to *overproduction*: producing beyond Q^* means units where $MC > WTP$, also destroying value. Q^* is the unique output that maximises TS.

Historical Episode: The 2008 Financial Crisis and the Allocative Cost of Too-Big-to-Fail.

The 2008 financial crisis offers a striking real-world illustration of deadweight loss — not from output restriction, but from *misallocation* of resources. Large US banks (Citigroup, Bank of America, AIG) were effectively insulated from market discipline by their implicit government guarantee: they knew they would be bailed out if insolvent. This guarantee is a *subsidy* — it lowers the banks' effective cost of risk-taking below its true social cost.

In the welfare framework: the banks produced “too much” risky lending (output beyond Q^* , where $MC_{\text{social}} > WTP_{\text{social}}$). The DWL triangle represents the value of resources poured into mortgage-backed securities and leveraged positions that generated negative social value. The subsequent bailout (TARP: \$700 billion authorised) was a redistribution — from taxpayers to bank creditors — but the DWL from the prior misallocation was not recovered.

The total economic cost of the crisis is estimated at between \$6 trillion and \$14 trillion in lost US GDP (Federal Reserve Bank of Dallas, 2013). This figure reflects the DWL from years of resource misallocation — a number that dwarfs the welfare triangles in any textbook diagram, but is generated by the same mechanism.

Lesson: DWL is not just a diagram concept. When markets systematically produce at the wrong output level — due to subsidies, externalities, or bailout guarantees — the welfare triangles scale up enormously.

Check Your Understanding

1. $D : P = 12 - Q$, $S : P = 2 + Q$. Find Q^* , P^* , CS, PS, TS. Verify $TS = \frac{1}{2}(10)(5) = 25$.
2. At $Q_1 = 3$: demand price = 9, supply price = 5. Compute the DWL triangle.
3. True or false: “If the government transfers \$10 from consumers to producers, total surplus falls.”

With the efficiency framework established, we can evaluate specific policies. International trade and tariffs are the most important applications.

8.5. Effects of Policy Interventions on Surplus

The domestic price of wheat is \$300/tonne. The world price is \$180. A policy advisor says opening trade is good for the country as a whole but bad for domestic farmers. How do we verify this precisely?

8.5.1. Opening Trade: The Import Case

When the world price $P_W < P_D$ (domestic price), opening trade:

- Domestic price falls from P_D to P_W .
- Domestic demand rises; domestic supply falls. Imports fill the gap.
- CS rises (consumers pay less); PS falls (domestic producers sell less).
- TS rises — the consumer gain exceeds the producer loss.

Gains from Trade: Opening imports when $P_W < P_D$ raises total surplus. The net gain (triangle BFE in Figure 8.4) equals $(1/2)(P_D - P_W)(Q_1 - Q_2)$ where Q_1 is post-trade demand and Q_2 is post-trade domestic supply.

In words: Free trade: consumers win big; domestic producers lose; net gain for society. Producers lobby for protection because their losses are concentrated; consumer gains are diffuse.

8.5.2. The Effect of a Tariff

A tariff t raises the domestic price from P_W to $P_R = P_W + t$.

Tariff Welfare Effects (starting from free trade):

- CS falls (price rises).
- PS rises (domestic producers get $P_R > P_W$).

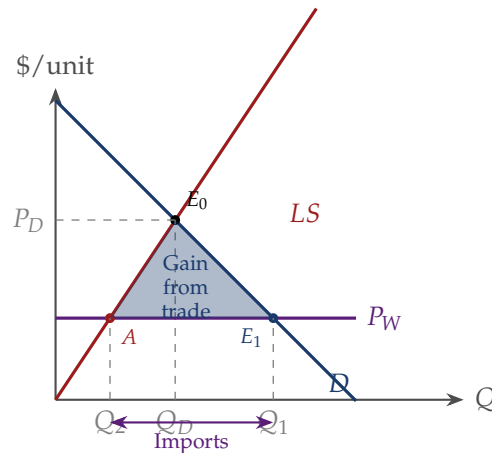


Figure 8.4. Opening trade when $P_W < P_D$. What to look for: At P_W , domestic supply falls to Q_2 and demand rises to Q_1 . Imports fill the gap. The gain-from-trade triangle (blue) is the net increase in TS: consumer gains exceed producer losses.

- Government collects revenue = $t \times \text{imports}$.
- Two DWL triangles remain: one from inefficient domestic production expansion; one from reduced consumption.
- Net: $\Delta TS < 0$. Tariffs reduce total surplus.

“Tariff revenue offsets the DWL.” Government revenue is a *transfer* (from consumers to the government) — still counted in TS. The two DWL triangles represent value that is *destroyed*, not transferred. Even if the government refunds every dollar of tariff revenue, the DWL triangles remain as lost value.

Worked Example. $D : P = 30 - Q$, $S : P = 6 + 0.5Q$, $P_W = \$10$, tariff $t = \$2$ so $P_R = \$12$.

At $P_W = 10$: $Q_s = 8$, $Q_d = 20$, imports = 12.

At $P_R = 12$: $Q_s^T = 12$, $Q_d^T = 18$, imports = 6.

$$\Delta CS = -\frac{1}{2}(P_R + P_W)(Q_d^T - Q_d^W) = -(P_R - P_W) \cdot \bar{Q}_d = -\$38.$$

$$\Delta PS = +\$20. \quad \text{Gov revenue} = 2 \times 6 = \$12.$$

$$\Delta TS = -38 + 20 + 12 = -\$6 \text{ (net DWL). } \checkmark$$

Try It: Tariff Welfare Decomposition. $D : P = 20 - Q$, $S : P = 2 + 0.5Q$, $P_W = \$6$, tariff $t = \$3$ so $P_R = \$9$.

- At $P_W = 6$: find Q_s and Q_d . How large are imports?
- At $P_R = 9$: find Q_s^T and Q_d^T . How large are imports now?
- Compute ΔCS , ΔPS , and government revenue.
- Compute the two DWL triangles. Verify $\Delta TS = \Delta CS + \Delta PS + \text{Gov rev}$.

Sketch the diagram showing the four areas: PS gain, CS loss, government revenue, and the two DWL triangles.

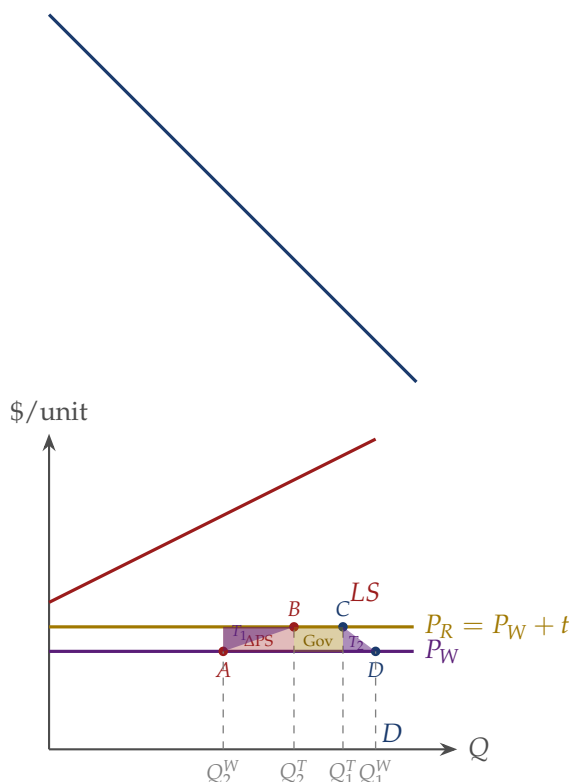


Figure 8.5. Tariff welfare effects. What to look for: A tariff raises the domestic price from P_W to $P_R = P_W + t$. Four areas result: a PS gain for domestic producers (red), government tariff revenue (gold), and two DWL triangles T_1 (production inefficiency) and T_2 (consumption reduction). CS falls by all four areas combined; the net effect on TS is $-(T_1 + T_2) < 0$.

Check Your Understanding

1. In the trade example: $P_D = \$14$, $P_W = \$10$, $Q_2 = 8$, $Q_1 = 20$. Compute the gain from trade as a triangle area.
2. After the tariff: why does DWL have *two* triangles rather than one?
3. A politician says “tariffs protect domestic jobs.” Use the surplus framework to evaluate this claim.

We have evaluated competitive and restricted-output outcomes. Now we introduce the most important non-competitive market structure: monopoly.

8.6. Monopoly

A firm is the only seller of a patented cancer drug. Unlike a wheat farmer, it can choose its price. How does it decide what price to charge — and why does it settle on something higher than marginal cost?

8.6.1. What Is a Monopoly?

Monopoly: A market with a single seller of a good for which there are no close substitutes. The firm *is* the market — it faces the entire downward-sloping market demand curve directly.

Barriers to entry keep rivals out. Two categories:

- *Technical:* ownership of a key resource; declining average cost over all output (natural monopoly); superior proprietary technology.
- *Legal:* patents, copyrights, exclusive government franchises.

Application: Pharma Patents and Monopoly Persistence.

A pharmaceutical company holds a patent on a cancer drug. During the patent period (typically 20 years), no rival may legally produce it. The company acts as a monopolist, setting price far above marginal cost. After patent expiry, generic manufacturers enter — supply increases, price falls toward MC, and DWL shrinks dramatically.

This is the intended trade-off in patent law: a temporary monopoly reward incentivises innovation, with competition restored afterwards. The question for policymakers is always: does the innovation benefit (dynamic efficiency) outweigh the DWL cost (static inefficiency)?

Data anchor. Empirical estimates of the Lerner Index for brand-name pharmaceuticals under patent protection typically range from $L \approx 0.5$ to $L \approx 0.9$, implying markups of 100%–900% above marginal production cost. For a widely cited example, Gleevec (imatinib) — a leukemia treatment — was priced at approximately \$92,000/year in the US while its estimated manufacturing cost was under \$1,000/year, implying $L > 0.98$ (estimates from the empirical health economics literature). After patent expiry and generic entry in 2016, prices fell by over 90%.

Monopolist = price maker (sets P).

Competitive firm = price taker (takes P as given).

8.6.2. Natural Monopoly and Regulation

Natural Monopoly: An industry in which a single firm can produce the market's total output at lower cost than two or more firms. This occurs when the average cost curve is *declining over the entire relevant range of output* — typically because fixed costs are very large and marginal costs are low (water networks, electricity grids, broadband infrastructure).

The Regulator's Dilemma.

Pricing rule	Efficiency	Viability
$P = MC$ (marginal-cost pricing)	Efficient (DWL = 0)	Loss-making ($P < AC$)
$P = AC$ (average-cost pricing)	Second-best (DWL > 0)	Break-even (zero profit)
Unregulated monopoly	Inefficient (DWL > 0)	Profitable

Most public utilities are regulated at average-cost pricing: a second-best outcome that preserves viability without requiring a subsidy.

In words: Natural monopoly regulation is a balancing act: push price toward MC to reduce DWL, but not so low that the firm exits.

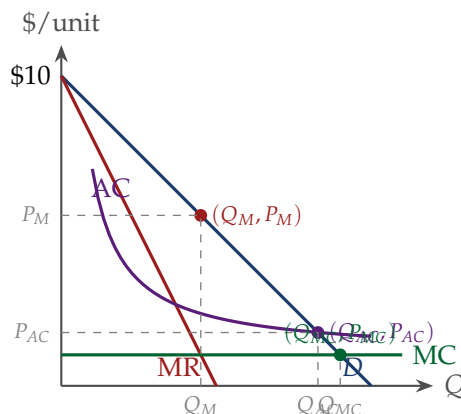


Figure 8.6. Natural monopoly: three regulatory outcomes. *What to look for:* The AC curve is declining throughout — a natural monopoly. Unregulated: (Q_M, P_M) produces lowest output and highest price (most DWL). AC pricing: (Q_{AC}, P_{AC}) — firm breaks even, more output, less DWL. MC pricing: (Q_{MC}, P_{MC}) — efficient output but $P < AC$, so the firm loses money and requires a subsidy.

Economic Insight.

Rent-Seeking. Monopoly profits create an incentive for firms to spend resources *obtaining or maintaining* their market power — through lobbying, patent litigation, regulatory capture, and predatory pricing against potential entrants. These expenditures are called **rent-seeking**. They are socially wasteful: they redistribute profit to the monopolist but create no new value. In the most extreme case, a firm might spend up to its entire expected monopoly profit in rent-seeking, so the true social cost of monopoly equals the DWL triangle *plus* the profit rectangle — a far larger number than the standard DWL calculation suggests.

8.6.3. Monopoly Profit Maximisation: The Two-Step Rule

The fundamental principle is the same as any firm: maximise profit by setting $MR = MC$. The critical difference from Week 7 is that the monopolist's $MR < P$ (it faces a downward-sloping demand curve).

Monopoly Two-Step Rule:

1. Find Q_M^* : Set $MR = MC$ and solve for output.
2. Find P_M^* : Read the demand curve at Q_M^* : $P_M^* = D(Q_M^*)$.

The monopolist **never** sets $P = MC$. Instead, $P_M^* = D(Q_M^*) > MC$.

In words: Step 1 finds the quantity. Step 2 finds the price. The demand curve determines what price the market will bear at the chosen quantity.

*Two-step: (1)
 $MR = MC \rightarrow Q_M^*$;
 (2) demand $\rightarrow P_M^*$.
 Never skip Step 2.*

8.6.4. Why $MR < P$ for a Monopolist

The monopolist must lower price on all units to sell one more:

$$MR = P + Q \frac{dP}{dQ} < P \quad (\text{since } dP/dQ < 0) \quad (8.4)$$

For linear inverse demand $P = a - bQ$:

$$MR = a - 2bQ \quad (8.5)$$

MR Curve Properties. For $P = a - bQ$: same y -intercept as demand; twice as steep (slope $-2b$ vs $-b$); hits zero at $Q = a/(2b)$ (half the demand choke quantity). At $MR = 0$, TR is maximised.

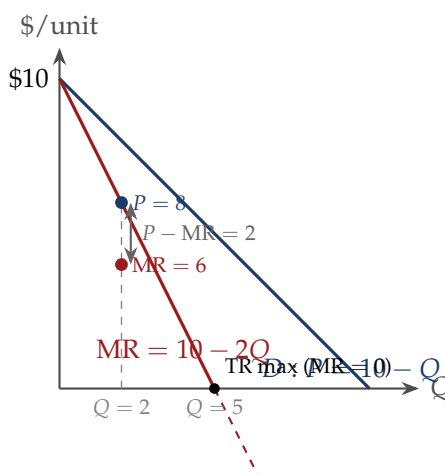


Figure 8.7. Why $MR < P$ for a monopolist. What to look for: At $Q = 2$, the demand curve gives $P = \$8$ but the MR curve gives $MR = \$6$. The gap ($P - MR = \2) represents the revenue lost on the first 2 units when price must fall to sell unit 3. The MR curve has the same y -intercept as demand ($\$10$) but twice the slope. TR is maximised where $MR = 0$ (at $Q = 5$, halfway to the choke quantity $Q = 10$).

A monopolist choosing between selling 3 units at \$7 or 4 units at \$6 faces the same fundamental tension as anyone who has ever sold a used car: charging more means selling to fewer people.

Confusing the MR curve with the demand curve. Both start at a on the price axis, but demand shows what consumers pay per unit; MR shows additional revenue from one more unit. Setting price = MR (or drawing the MR curve as the demand curve) is a frequent error that gives the wrong Q_M^* .

Solved Example 8.1: $P = 10 - Q$, $MC = AC = \$2$.

$MR = 10 - 2Q$. Step 1: $10 - 2Q = 2 \Rightarrow Q_M^* = 4$. Step 2: $P_M^* = 10 - 4 = \$6$. Profit = $4(6 - 2) = \$16$.

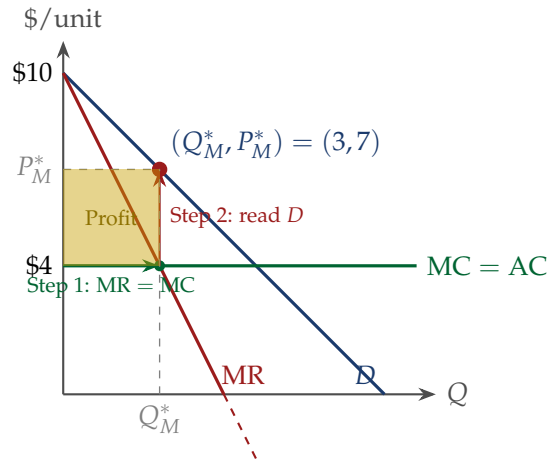


Figure 8.8. Monopoly equilibrium: $D : P = 10 - Q$, $MC = \$4$. What to look for: Step 1 — find $Q_M^* = 3$ where $MR = MC$. Step 2 — read the demand curve at Q_M^* to get $P_M^* = \$7$. Profit (yellow) $= Q_M^*[P_M^* - AC] = 3(7 - 4) = \9 . The monopolist does *not* set $P = MC$.

Solved Example 8.2: $Q = 70 - P$, $MC = AC = \$6$.

Inverse demand: $P = 70 - Q$. $MR = 70 - 2Q$. Step 1: $70 - 2Q = 6 \Rightarrow Q_M^* = 32$. Step 2: $P_M^* = 70 - 32 = \$38$. Profit $= 32(38 - 6) = \$1,024$.

8.6.5. The Lerner Index: Measuring Market Power

Lerner Index:

$$L = \frac{P - MC}{P} = -\frac{1}{\varepsilon_d} \quad (8.6)$$

$L = 0$ for a competitive firm ($P = MC$). $L \rightarrow 1$ as market power becomes extreme ($MC/P \rightarrow 0$). At the monopoly optimum: $L = -1/\varepsilon_d$ directly relates markup to demand elasticity.

In words: Lerner Index = fraction of price that is markup above MC. Higher L = more market power.

Check Your Understanding

1. $D : P = 50 - 2Q$, $TC = 10Q + 100$. Apply the two-step rule: find Q_M^* , P_M^* , and profit.
2. For the benchmark $D : P = 10 - Q$, $MC = 4$: verify $Q_M^* = 3$, $P_M^* = 7$. Compute the Lerner Index.
3. A monopolist has $L = 0.4$ and $P = \$50$. What is MC ?

In plain English:
 $L = 0.5$ means half the price is markup. $L = 0$ means competitive pricing.

The monopoly equilibrium delivers positive profit but restricts output below the competitive level. The next section quantifies the welfare cost of this restriction.

8.7. Why Are Monopolies Undesirable? A Welfare Comparison

A monopolist charges \$7 where a competitive market would charge \$4, producing 3 units instead of 6. Con-

sumers pay more and buy less. But is this just a transfer from consumers to the firm — or is there a genuine loss to society?

8.7.1. The Key Welfare Result

Using our benchmark ($D : P = 10 - Q$, $MC = AC = 4$, verified above):

Competitive vs. Monopoly: Verified Numbers

Scenario	CS	PS/Profit	TS	DWL
Perfect competition ($Q_{PC} = 6, P_{PC} = 4$)	18	0	18	0
Monopoly ($Q_M = 3, P_M = 7$)	4.5	9	13.5	4.5
Change	-13.5	+9	-4.5	+4.5

Verify: $CS_{PC} = (1/2)(10 - 4)(6) = 18$. $CS_M = (1/2)(10 - 7)(3) = 4.5$. $Profit_M = (7 - 4)(3) = 9$. $DWL = (1/2)(7 - 4)(6 - 3) = 4.5$. ✓

The rectangle $ABDC$ (\$9 = monopoly profit) is a *transfer* from consumers to the firm. The triangle CDE (\$4.5 = DWL) is *destroyed* — not captured by anyone. Total surplus falls by \$4.5.

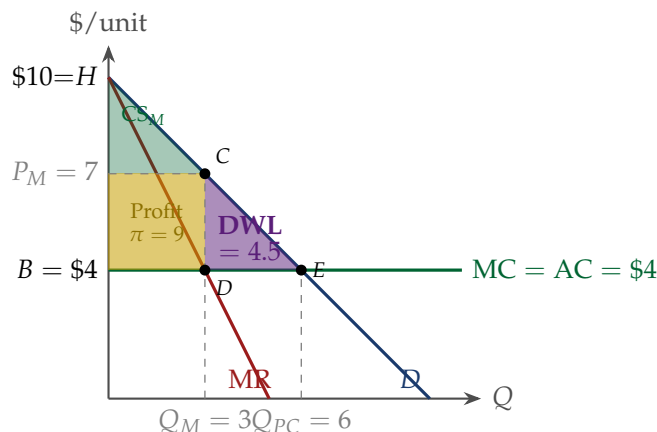


Figure 8.9. Monopoly DWL: benchmark $D : P = 10 - Q$, $MC = \$4$. What to look for: Yellow rectangle = monopoly profit (transfer from consumers, \$9). Purple triangle CDE = DWL (\$4.5) — value destroyed. TS falls from \$18 to \$13.5. Not all CS loss is DWL: the rectangle is a transfer; only the triangle is genuinely destroyed.

Why This Matters: Antitrust. When the US Department of Justice or the European Commission evaluates a proposed merger, they use exactly the framework in Figure 8.9. They estimate whether the resulting firm would have enough market power to raise prices significantly above MC — and whether the DWL triangle would be large enough to harm consumers and society. Blocking a merger is not just about “fairness” — it is about preserving the efficiency of markets.

Check Your Understanding

1. Using the benchmark: verify $DWL = (1/2)(P_M - MC)(Q_{PC} - Q_M) = (1/2)(3)(3) = 4.5$. ✓
2. “Monopoly just transfers surplus from consumers to the firm.” True or false? Explain carefully.
3. A monopolist has $L = 0.5$. What does this imply about demand elasticity at the optimum?

Two worked examples consolidate the two-step rule with different cost structures.

8.8. Worked Examples: Monopoly Profit Maximisation

Two worked examples show the two-step rule with constant and variable MC. The key numbers (Q_M^* , P_M^* , π , DWL) are all verified.

Example 8.3: $Q = 70 - P$, $MC = AC = \$6$.

Inverse demand: $P = 70 - Q$. $TR = (70 - Q)Q = 70Q - Q^2$. $MR = 70 - 2Q$.

Step 1: $70 - 2Q = 6 \Rightarrow Q_M^* = 32$. Step 2: $P_M^* = 70 - 32 = \$38$.

Profit = $TR - TC = 32(38) - 32(6) = 1,216 - 192 = \$1,024$.

Competitive baseline: $P = MC \Rightarrow 70 - Q = 6 \Rightarrow Q_{PC} = 64$, $P_{PC} = \$6$.

$DWL = (1/2)(P_M^* - MC)(Q_{PC} - Q_M^*) = (1/2)(32)(32) = \512 .

Example 8.4: $Q = 70 - P$, $TC = 0.25Q^2 - 5Q + 300$.

$MC = 0.5Q - 5$. $MR = 70 - 2Q$.

Step 1: $70 - 2Q = 0.5Q - 5 \Rightarrow 75 = 2.5Q \Rightarrow Q_M^* = 30$. Step 2: $P_M^* = 70 - 30 = \$40$.

$TC(30) = 0.25(900) - 150 + 300 = 225 - 150 + 300 = \375 . Profit = $30(40) - 375 = 1,200 - 375 = \825 .

Try It. For Example 8.3: using $P = MC = \$6$ at $Q_{PC} = 64$, compute $CS_{PC} = (1/2)(70 - 6)(64) = \$2,048$. Under monopoly: $CS_M = (1/2)(70 - 38)(32) = \512 . Verify $DWL = CS_{PC} + 0 - (CS_M + \text{profit}) = 2,048 - (512 + 1,024) = \512 . ✓

We have analysed the single-price monopolist. Now we consider a monopolist sophisticated enough to charge different prices to different customers — price discrimination.

8.9. Price Discrimination

Airlines charge business travellers three times what leisure travellers pay for the same seat. Is this just greedy — or is there economic logic, and what are the welfare implications?

8.9.1. Definition and Conditions

Price Discrimination: Charging different prices for identical (or near-identical) units of output to different buyers.

Two necessary conditions:

- (1) The seller must have **market power** (cannot price discriminate as a price taker).
- (2) The seller must be able to prevent **arbitrage** — low-price buyers reselling to high-price buyers.

In words: Price discrimination works only if the firm can keep markets separate. Without this, the price differential collapses.

8.9.2. The Efficiency–Equity Trade-Off: The Central Insight

Three Market Structures: The Complete Welfare Picture

Scenario	CS	PS/Profit	TS	DWL
Perfect competition	Max	0	Max	0
Uniform-price monopoly	Medium	Medium	Less than max	> 0
Perfect price discrimination (PPD)	0	Max	Max	0

Key insight: Perfect competition and PPD tie on *efficiency* (TS is maximised, DWL = 0) but are polar opposites on *equity*. Competition gives all surplus to consumers; PPD gives all surplus to the producer.

Efficiency $\neq$ Equity. This table crystallises Week 8’s central lesson. Perfect price discrimination is socially efficient (no DWL, output at Q_{PC}) but deeply inequitable (consumers get nothing). Competitive markets are both efficient and equitable. Uniform monopoly is neither. Policymakers must decide which dimension they prioritise.

8.9.3. Perfect Price Discrimination (First-Degree)

Perfect Price Discrimination (PPD): Each consumer is charged exactly their willingness to pay. Every unit is sold at the price indicated by the demand curve. Since no consumer retains any surplus, CS = 0.

The monopolist sells units as long as $WTP \geq MC$, reaching output Q_{PC} . **No DWL.** All surplus accrues to the monopolist as profit.

In words: PPD is efficient (right quantity produced) but inequitable (all value goes to the seller). The pie is as big as possible — but the seller eats it all.

8.9.4. Second-Degree Price Discrimination: Versioning and Quantity Discounts

Second-Degree Price Discrimination: Charging different prices based on the *quantity purchased* or the *version of the product chosen* — not on the identity of the buyer. The firm cannot observe individual WTP directly; instead, it designs a menu of options that induces buyers to self-select.

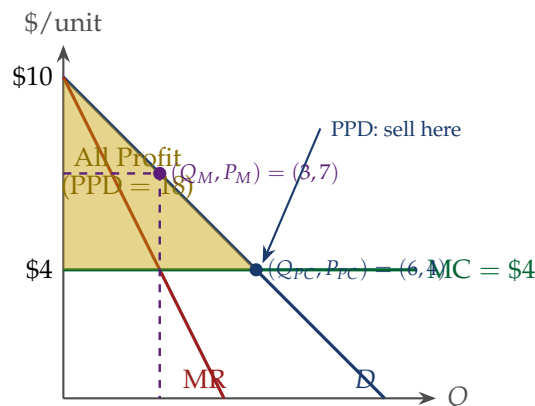


Figure 8.10. Perfect price discrimination (PPD). What to look for: The entire triangle under demand above MC becomes profit (\$18). No DWL — output reaches $Q_{PC} = 6$. No CS — consumers pay exactly their WTP. Under standard monopoly (dashed), profit = \$9, CS = \$4.5, DWL = \$4.5.

Two common forms:

- **Quantity discounts:** lower per-unit price for larger purchases (wholesale clubs, utility block pricing).
- **Versioning:** offering multiple product tiers (basic/premium) at different price points, so high-WTP consumers self-select into the expensive version.

In words: Second-degree PD extracts more surplus than uniform pricing without requiring individual buyer identification. The menu does the work.

Application: Software Versioning as Second-Degree Price Discrimination.

Microsoft Office sells “Home” and “Business” versions of the same underlying software at \$100 and \$300 respectively. The Business version adds a few features (Outlook, Teams integration) but the marginal cost of those features is nearly zero. The price difference is not driven by cost — it is driven by self-selection: price-sensitive home users choose the cheaper tier, while business users (whose employers pay and who value seamless corporate integration) choose the premium tier.

The key insight is that Microsoft does *not* need to identify whether you are a home user or a business user. The product design does the sorting automatically. This is second-degree PD: a menu of versions that allows buyers to reveal their own willingness to pay through the tier they choose.

Welfare note: second-degree PD typically increases total output relative to uniform monopoly pricing (more consumers are served at the lower tier) and thus *reduces* DWL — though CS is still lower than under perfect competition.

Check Your Understanding

1. A cinema charges adults \$14 and students \$9 for the same film. What type of price discrimination is this? What arbitrage problem must the cinema solve?
2. Microsoft sells Office Home (\$100) and Office Business (\$300). The marginal cost

of the extra Business features is nearly zero. What type of PD is this? What determines which tier a user selects?

3. True or false: "Second-degree price discrimination always increases CS relative to uniform monopoly pricing." Explain.

8.9.5. Market Separation (Third-Degree Price Discrimination)

Market Separation: The monopolist divides customers into identifiable groups with distinct demand curves and acts as a separate monopolist in each market.

Optimal rule: In each market i , set $MR_i = MC$ to find Q_i^* , then read demand for P_i^* .

Pricing rule: Charge a *higher price* in the market with *less elastic* demand.

In words: Higher price in the captive market (less elastic), lower price in the competitive market (more elastic).

Application: Airline Price Discrimination.

Airlines are textbook market separators. They divide passengers into:

- **Business travellers:** less elastic demand (employer pays, short notice, must travel). Higher price.
- **Leisure travellers:** more elastic demand (flexible, plan ahead, can substitute). Lower price.

Arbitrage prevention: tickets are non-transferable, require ID, and have advance-purchase restrictions. A leisure traveller cannot resell their seat to a last-minute business traveller.

Numbers: A business fare might be \$1,200; the same route on the same day in economy might cost \$300 booked two weeks ahead. The airline applies the two-step rule independently to each market segment, charging each what demand will bear.

Two-Market Worked Example.

$D_1 : Q_1 = 100 - P_1$; $D_2 : Q_2 = 100 - 2P_2$; $MC = \$20$.

Market 1: $P_1 = 100 - Q_1$. $MR_1 = 100 - 2Q_1 = 20 \Rightarrow Q_1^* = 40$. $P_1^* = \$60$. $\pi_1 = (60 - 20)(40) = \$1,600$.

Market 2: $P_2 = 50 - Q_2/2$. $MR_2 = 50 - Q_2 = 20 \Rightarrow Q_2^* = 30$. $P_2^* = \$35$. $\pi_2 = (35 - 20)(30) = \450 .

Total: $\pi_{disc} = \$2,050$.

Without discrimination (uniform price): $Q = 200 - 3P$ (total demand). $MR = (200 - 2Q)/3$. Setting $MR = MC = 20$: $Q^* = 70$, $P^* \approx \$43.33$. $\pi_{uniform} \approx \$1,633$.

Price discrimination adds $\approx \$417$ in profit. Market 1 (less elastic demand) pays the higher price. ✓

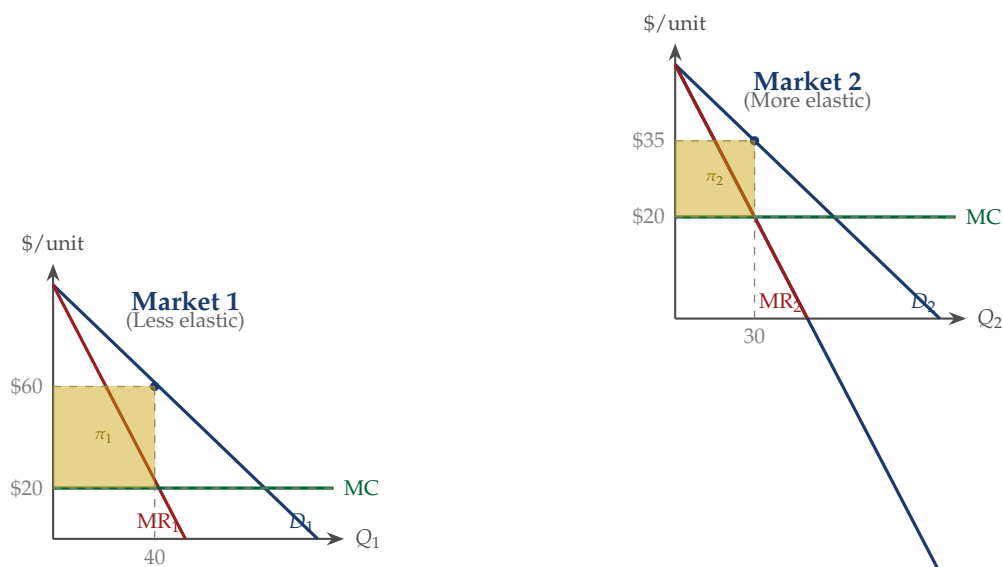


Figure 8.11. Third-degree price discrimination: two-market solution. *What to look for:* In each market the monopolist applies the two-step rule independently: set $MR_i = MC$, then read the demand curve. Market 1 (less elastic) bears the higher price (\$60); Market 2 (more elastic) bears the lower price (\$35). Profit rectangles are shown in yellow. The key condition is $MR_1 = MR_2 = MC = \$20$ at the optimum.

The Economist's Lens: How Spotify Uses Price Discrimination to Expand the Market

Spotify charges \$11.99/month for an individual premium subscription, \$5.99/month for students, and offers a free ad-supported tier. This is a deliberate three-tier price discrimination strategy combining elements of second-degree (versioning: free vs. paid) and third-degree (market separation: students vs. non-students).

The welfare logic:

- **Free tier:** captures users with WTP near zero. Generates ad revenue; crucially, serves as the screening device that separates low-WTP from high-WTP users.
- **Student tier:** students have elastic demand (limited income, many substitutes including piracy). Lower price expands the market; arbitrage is prevented by requiring university email verification.
- **Standard tier:** less price-sensitive adults with stable incomes pay the full price.

Why it's efficient relative to uniform pricing: Without price discrimination, Spotify would set a single price that excludes low-WTP listeners entirely. The multi-tier structure allows more consumers to be served — output increases, DWL falls — while the firm captures more surplus than under a uniform price.

Concept illustrated: Market separation (third-degree PD) + versioning (second-degree PD) working simultaneously; arbitrage prevention as the enabling mechanism; PD as a welfare improvement relative to single-price monopoly.

Check Your Understanding

1. In the two-market example, verify that Market 1 has less elastic demand at the optimum. (Compute $|\varepsilon_{d1}|$ at $Q_1^* = 40$, $P_1^* = 60$ using $Q_1 = 100 - P_1$.)
2. Under PPD with our benchmark ($D : P = 10 - Q$, $MC = 4$): compute profit. Verify $TS = 18 =$ competitive TS . ✓
3. A software company charges universities \$50 and businesses \$500 for the same programme. What conditions must hold for this to be sustainable?

The content is complete. Before the practice questions, review the exam mistakes, back-matter reference material, and revision summary.

Where Does This Take Us?

- **Week 9 — Oligopoly and Game Theory:** Between perfect competition and monopoly lies oligopoly — a few large firms. Game theory (Nash equilibrium, prisoner's dilemma) models how firms interact strategically. The welfare tools from Week 8 apply: oligopoly DWL lies between competitive zero and monopoly maximum.
- **Trade policy:** The tariff DWL framework from Section 8.5 underpins all trade policy analysis. Specific tariffs, ad-valorem tariffs, quotas, and export subsidies all create DWL triangles analysed the same way.
- **Natural monopoly regulation:** The tension between marginal-cost pricing (efficient but loss-making) and average-cost pricing (second-best but viable) introduced in Section 8.6 is a direct application this chapter. Regulators use this trade-off when setting utility prices.
- **The Lerner Index** $L = -1/\varepsilon_d$ connects Week 8 market power analysis to Week 4 elasticity. It is also used in antitrust litigation to measure market power empirically.
- **Second-degree price discrimination and versioning** appear throughout digital markets (software, streaming, cloud services) and will recur in the context of platform economics and two-sided markets in advanced courses.

Five Common Exam Mistakes — and How to Avoid Them

1. **Treating all CS loss under monopoly as DWL.** The monopoly CS loss has two parts: the rectangle $ABDC$ (transferred to the monopolist as profit) and the triangle CDE (genuine DWL). Only the triangle is deadweight loss — the rectangle is a redistribution, not a destruction. Always decompose CS loss into transfer and DWL.
2. **Forgetting to read the demand curve for P_M^* .** The two-step rule requires Step 2: after finding Q_M^* from $MR = MC$, read the demand curve at Q_M^* to find P_M^* . Many students stop at Step 1 and report $P_M^* = MC$. This gives the competitive price, not the monopoly price.
3. **Confusing the MR curve with the demand curve.** Both have the same y -intercept. But the MR curve is twice as steep. Setting $P = MR$ instead of $MR = MC$ gives the wrong Q_M^* .
4. **Claiming tariff revenue offsets DWL.** Government tariff revenue is a transfer (from consumers to the government) and is included in TS . The two DWL triangles represent value

that is destroyed even after counting government revenue. DWL and tariff revenue are separate components.

5. **Saying perfect price discrimination is “better” without qualification.** PPD achieves the efficient quantity and eliminates DWL — but consumer surplus is zero. It is efficient but not equitable. A correct answer must acknowledge both dimensions: PPD ties with competition on TS but is the worst possible outcome for consumers.

Chapter at a Glance — Week 8: Welfare, Monopoly & Price Discrimination

Your one-page revision guide.

Key Equations

$CS = WTP - P$	CS definition	p. 5
$PS = P - MC$	PS definition	p. 7
$TS = CS + PS$	total surplus	p. 8
$DWL = TS^* - TS_{\text{actual}}$	deadweight loss	p. 8
$MR = a - 2bQ$ for $P = a - bQ$	monopoly MR	p. 14
$L = (P - MC)/P = -1/\varepsilon_d$	Lerner Index	p. 15

The Three-Scenario Table (benchmark: $D : P = 10 - Q$, $MC = 4$)

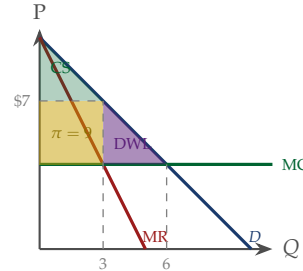
	CS	Profit	TS	DWL
Competition	18	0	18	0
Monopoly	4.5	9	13.5	4.5
PPD	0	18	18	0

PC and PPD tie on efficiency; PC wins on equity.

New Concepts Introduced This Week

Concept	Section
Consumer surplus (CS)	§8.2
Producer surplus (PS)	§8.3
Economic efficiency; DWL	§8.4
Gains from trade; tariff DWL	§8.5
Monopoly; barriers to entry	§8.6
Natural monopoly; regulation	§8.6
Rent-seeking	§8.6
Two-step monopoly rule	§8.6
Lerner Index	§8.6
Monopoly DWL vs. transfer	§8.7
Price discrimination (1st, 2nd, 3rd)	§8.9

Monopoly Diagram



LO Checklist

- LO1: Compute CS and PS ✓
- LO2: Efficiency as TS max; DWL ✓
- LO3: Trade and tariff welfare effects ✓
- LO4: Two-step monopoly rule ✓
- LO5: Monopoly DWL; Lerner Index ✓
- LO6: PPD and market separation ✓

The thread: $CS + PS = TS \rightarrow$ competitive equilibrium maximises $TS \rightarrow$ output restriction creates $DWL \rightarrow$ trade opens: $CS \uparrow$, $PS \downarrow$, net $TS \uparrow \rightarrow$ tariff creates two DWL triangles $\rightarrow$ monopoly: $MR = MC$ (two-step), $DWL = \text{triangle } CDE \rightarrow$ PPD: efficient but $CS = 0$. Central lesson: efficiency and equity are distinct — a market can maximise the pie while distributing it very unequally.

8.10. Key Terms

Consumer Surplus (CS)

WTP $- P$; area under demand curve and above price. *Used throughout the course.*

Producer Surplus (PS)

Price $- MC$ (willingness to sell); area above supply curve below price. Note: PS $\neq$ profit when fixed costs are present.

Total Surplus (TS)

CS + PS; the monetary welfare metric for comparing outcomes.

Economic Efficiency

TS is maximised; no mutually beneficial trades remain unexploited.

Deadweight Loss (DWL)

Reduction in TS from a distortion; value destroyed, not transferred.

Gains from Trade

Rise in TS when opening imports at $P_W < P_D$.

Tariff

Tax on imports; raises domestic price above world price; creates two DWL triangles.

Monopoly

Single seller with market power; faces downward-sloping demand; $MR < P$.

Barriers to Entry

Technical or legal obstacles preventing competitors from entering.

Natural Monopoly

Declining AC over entire market output range; one firm is cheapest. Regulated at

average-cost pricing as a second-best solution.

Rent-Seeking

Resources spent to obtain or maintain monopoly power (lobbying, litigation); social cost above and beyond the standard DWL triangle.

Two-Step Rule

Monopoly optimum: (1) set $MR = MC$ to find Q_M^* ; (2) read demand for P_M^* .

Monopoly DWL

Triangle CDE ; value of trades between Q_M and Q_{PC} that do not occur.

Lerner Index

$L = (P - MC)/P = -1/\epsilon_d$; measure of market power; $L = 0$ (competition), $L \rightarrow 1$ (extreme power). *Used throughout the course.*

Price Discrimination

Charging different prices for identical units; requires market power and arbitrage prevention.

Perfect Price Discrimination (PPD)

Each consumer charged their exact WTP; CS = 0; DWL = 0; efficient but inequitable.

Second-Degree Price Discrimination

Prices vary by quantity or product version; buyers self-select (versioning, quantity discounts).

Market Separation

Third-degree PD: separate monopoly pricing in identifiable groups; higher price in less elastic market.

Recommended Videos

Four videos selected for clarity and direct relevance to this chapter.

1. Consumer and Producer Surplus (*Jacob Clifford, ACDC Economics*) ~ 9 min

<https://www.youtube.com/watch?v=CtCfwMmzOBU>

Derives CS and PS graphically for linear supply and demand, explains the total surplus triangle, and shows how a price change affects each component. Reinforces Sections 8.2–8.4 and ideal before Level 1–2 surplus questions.

2. Taxation and Deadweight Loss (*Khan Academy*) ~ 11 min

<https://www.youtube.com/watch?v=NuLlNAdrom4>

Covers how a per-unit tax creates a tax wedge and generates deadweight loss, with clear triangle diagrams. Directly reinforces Sections 8.4–8.5. Recommended before Level 2 tax incidence questions.

3. Monopoly and Deadweight Loss (*Jacob Clifford, ACDC Economics*) ~ 10 min

<https://www.youtube.com/watch?v=YNcPxPz9fng>

Two-step rule, profit rectangle, DWL triangle, and the CS/PS/TS comparison between monopoly and competition. Reinforces Sections 8.6–8.7 and is recommended before Level 2–3 monopoly problems.

4. Price Discrimination (*Jacob Clifford, ACDC Economics*) ~ 9 min

<https://www.youtube.com/watch?v=s3wFJHIuJPs>

Covers the three degrees of price discrimination, conditions for profitable price discrimination, and the efficiency–equity trade-off. Reinforces Section 8.9 and the comparison table.

8.11. Practice Questions

How to use these questions. Difficulty: ★ easy ★★ intermediate ★★★ challenging.
Attempt every question before consulting the solutions.

Level 1: Conceptual and Short-Answer

- ★ **[CS and PS — Concepts]** Define consumer surplus and producer surplus. Which curve represents consumers' willingness to pay? Which represents producers' willingness to sell? Why is total surplus a useful metric for comparing markets that trade completely different goods?
- ★ **[Efficiency and Deadweight Loss]**
 - (a) State the two equivalent definitions of economic efficiency from this chapter.
 - (b) Why does producing *fewer* than Q^* units create a DWL? Who loses, and does anyone gain?
 - (c) Why does producing *more* than Q^* units also create a DWL?
- ★ **[Gains from Trade]** A country's domestic wheat price is \$300/tonne and the world price is \$180/tonne.
 - (a) Will the country import or export? Explain.
 - (b) What happens to domestic CS, PS, and TS when trade opens?
 - (c) Which group is harmed? Why might they lobby against free trade?
 - (d) Is the country as a whole better or worse off? Justify with the total surplus concept.
- ★ **[Why Monopolies Are Undesirable]**
 - (a) Why does a monopolist charge a higher price and produce less than a competitive market?
 - (b) A student argues: "Monopoly just transfers surplus from consumers to the firm — it's a distributional issue, not an efficiency problem." Is this correct?
 - (c) Beyond the DWL triangle, describe one additional social cost of monopoly.
- ★ **[Price Discrimination — Concepts]**
 - (a) Define price discrimination. What market condition is necessary?
 - (b) Why must a firm prevent arbitrage for price discrimination to work?
 - (c) Explain the difference between perfect (first-degree) and market separation (third-degree) price discrimination.
 - (d) Under PPD, $CS = 0$ and $DWL = 0$. Does this mean PPD is socially desirable?

★ [The Lerner Index]

- (a) Explain what the Lerner Index measures. Why is it zero for a competitive firm?
- (b) A firm has $L = 0.5$. What fraction of the price is markup above MC?
- (c) Using $L = -1/\varepsilon_d$: why does a firm in a market with inelastic demand ($|\varepsilon_d| = 1.2$) set a higher markup than one with elastic demand ($|\varepsilon_d| = 4$)?
- (d) Can the Lerner Index exceed 1? Why or why not?

★ [Equity vs. Efficiency] Consider: (i) perfect competition, (ii) uniform-price monopoly, (iii) perfect price discrimination.

- (a) Rank by total surplus. Explain why two scenarios tie.
- (b) Rank by consumer surplus. Which is best and worst for consumers?
- (c) A policymaker cares equally about efficiency and consumer welfare. She chooses between uniform monopoly and PPD. Which would you advise, and why?

★ [Explain to a Friend — Level 1] *No formulas.* Your friend says: “Monopoly profits are just money going from consumers to the company — society doesn’t actually lose anything.” Write 3–4 plain-English sentences explaining why this is wrong and what the deadweight loss actually represents.

Level 2: Intermediate Problems

★★ [Computing CS, PS, TS from Linear S&D] $D : P^d = 12 - Q$, $S : P^s = 2 + Q$.

- (a) Find P^* and Q^* .
- (b) Calculate CS, PS, and TS.
- (c) Government imposes price ceiling $\bar{P} = 4$. Compute CS, PS, DWL. Who gains and who loses?
- (d) Verify $DWL = TS_{PC} - TS_{ceiling}$.

★★ [Import Case: Full Welfare Analysis] $D : P^d = 30 - Q$, $S : P^s = 6 + 0.5Q$, $P_W = \$10$.

- (a) Find the domestic equilibrium (P_D, Q_D) and compute CS_D , PS_D , TS_D .
- (b) After trade opens: find domestic Q_s^W , Q_d^W , and imports. Compute CS, PS, TS.
- (c) Compute ΔCS , ΔPS , ΔTS . Identify the gain-from-trade triangle.

★★ [Tariff Analysis] Continue from Q10. Tariff $t = \$2$, $P_R = \$12$.

- (a) Domestic Q_s^T , Q_d^T , imports after tariff.
- (b) Compute CS, PS, and government revenue at $P_R = 12$.
- (c) Compute DWL. Identify the two DWL triangles.

(d) Who gains and who loses from the tariff relative to free trade?

★★ [Monopoly: Basic Profit Maximisation] $P = 50 - 2Q$, $TC = 10Q + 100$.

- (a) Derive MR and MC.
- (b) Find Q_M^* , P_M^* , and profit using the two-step rule.
- (c) Competitive baseline: find Q_{PC} , P_{PC} .
- (d) Compute CS, PS, TS under (i) monopoly, (ii) perfect competition. Calculate DWL.

★★ [Monopoly with Quadratic Costs] $P = 60 - Q$, $TC = Q^2 + 4Q + 20$.

- (a) Derive MR, MC, AC.
- (b) Find (Q_M^*, P_M^*) and profit.
- (c) Find competitive equilibrium.
- (d) Show $DWL = TS_{PC} - TS_M$ expressed as a triangle.

★★ [Monopoly Markup and Lerner Index] $Q = 200 - 4P$, $MC = \$20$.

- (a) Find Q_M^* , P_M^* , and profit.
- (b) Compute the Lerner Index. Interpret.
- (c) Verify $L = -1/\varepsilon_d$ at the monopoly price.

★★ [Perfect Price Discrimination] $P = 80 - Q$, $MC = AC = \$20$.

- (a) Standard monopoly: Q_M^* , P_M^* , profit, CS, DWL.
- (b) Under PPD: output, profit, CS, TS.
- (c) Compare TS under (i) competition, (ii) monopoly, (iii) PPD.
- (d) Does PPD benefit consumers? Explain.

★★ [Data-Based Question] The table shows a monopolist's price–quantity data.

Q	P	TR	MR	MC
0	100	0	—	—
2	90	180		20
4	80	320		20
6	70	420		20
8	60	480		20
10	50	500		20
12	40	480		20

- (a) Fill in the MR column (computed between consecutive rows, per unit).
- (b) Identify the profit-maximising output. Compute monopoly profit.
- (c) Compute CS and TS at the monopoly equilibrium.
- (d) Find the competitive output (where $P = MC = \$20$). Compute DWL.
- (e) At which output is TR maximised? What is MR there? What does this imply about demand elasticity?
- (f) Under PPD: compute the monopolist's profit and verify $DWL = 0$.

★★ [Market Separation] $D_1 : Q_1 = 120 - 2P_1$, $D_2 : Q_2 = 80 - P_2$, $MC = \$10$.

- (a) Derive inverse demand and MR in each market.
- (b) Find (Q_1^*, P_1^*) and (Q_2^*, P_2^*) under price discrimination.
- (c) Total profit under discrimination?
- (d) Optimal uniform price and profit?
- (e) How much does discrimination increase profit?

★★ [Explain to a Friend — Level 2] Your friend says: “I understand that a monopolist restricts output. But why does that create a deadweight loss? Surely if the monopolist is making profit, that value still exists in the economy.” Write a 4–5 sentence explanation using the concept of mutually beneficial trades.

Level 3: Advanced Problems

★★★ [Full Welfare Comparison Table] $Q^d = 100 - P$, $MC = \$20$ (constant).

- (a) Competitive equilibrium: CS, PS, TS.
- (b) Monopoly: (Q_M^*, P_M^*) , profit, CS, DWL.
- (c) PPD: output, profit, CS, TS.
- (d) Fill in the complete welfare table. Comment on equity vs. efficiency.

★★★ [Monopoly: General Demand and Comparative Statics] $P = a - bQ$, $MC = c$.

- (a) Derive $Q_M^*(a, b, c)$ and $P_M^*(a, b, c)$.
- (b) Derive maximised profit $\pi^*(a, b, c)$.
- (c) Find $\partial Q_M^* / \partial c$ and $\partial P_M^* / \partial c$. Interpret.
- (d) Show monopoly markup $P_M^* - c = (a - c) / 2$.
- (e) Derive DWL as a function of a, b, c . Show DWL rises as b decreases.

★★★ [Market Separation: General Case and Elasticity Rule] $P_i = a_i - b_i Q_i$, $MC = c$.

- (a) Derive P_1^* and P_2^* under discrimination.
- (b) Show $P_1^* > P_2^*$ iff $|\varepsilon_{d1}| < |\varepsilon_{d2}|$ at respective equilibria.
- (c) Verify the inverse elasticity rule: $(P_i - MC)/P_i = -1/\varepsilon_{di}$.
- (d) With $a_1 = 100, b_1 = 2, a_2 = 80, b_2 = 1, c = 10$: find optimum under discrimination and compare to uniform pricing.

★★★ [Natural Monopoly and Regulation] $P = 100 - Q$, $TC = 500 + 5Q$ (so $MC = 5$, $AC = 500/Q + 5$, always declining).

- (a) Unregulated monopoly: (Q_M^*, P_M^*) and profit.
- (b) Marginal-cost pricing ($P = MC = 5$): find Q , CS , firm profit. Is there a loss?
- (c) Average-cost pricing ($P = AC$): find Q , CS , TS .
- (d) Compare the three outcomes in a table (Q , P , profit, CS , TS). Explain the regulator's trade-off.

★★★ [Monopoly with Non-Constant MC] $P = 100 - Q$, $TC = \frac{1}{3}Q^3 - 5Q^2 + 40Q + 50$.

- (a) Derive MC . Find the output minimising MC .
- (b) Set $MR = MC$. Find candidates for Q_M^* . Apply the SOC.
- (c) Find P_M^* and profit. Find Q_{PC} ; verify $Q_{PC} > Q_M^*$.

★★★ [Full Welfare Comparison: Trade, Monopoly, Price Discrimination] $D : P = 200 - 2Q$, single domestic producer with $MC = AC = \$20$.

- (a) Competitive baseline: CS , PS , TS .
- (b) Monopoly: (Q_M^*, P_M^*) , profit, CS , DWL .
- (c) Free trade at $P_W = \$60$: CS , PS , TS . Tariff $t = \$40$: CS , PS , government revenue, TS , DWL .
- (d) Price discrimination with $D_A : Q_A = 60 - P_A$ and $D_B : Q_B = 40 - 0.5P_B$ ($MC = \$20$): find (Q_A^*, P_A^*) , (Q_B^*, P_B^*) , total profit.
- (e) Rank all scenarios by TS . Comment on best/worst for consumers and the producer.

★★★ [Explain to a Friend — Level 3] Your friend, a business student, says: "If price discrimination is efficient (no DWL), then regulators should allow it wherever possible. Who cares if consumers get zero surplus?" Write a 5–6 sentence response addressing: (a) why PPD achieves efficiency; (b) why "zero consumer surplus" is still a serious concern; (c) why policymakers might prefer uniform-price monopoly over PPD ; and (d) what an alternative policy might achieve both efficiency and consumer protection.

Full solutions follow in the solutions document. Attempt all questions independently before consulting them.